

Program : Diploma in Automation and Robotics	
Course Code: 6332B	Course Title: Environmental Impact Assessment
Semester: 3	Credits: 4
Course Category : Program core	
Periods per week: 4(L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To introduce the methodologies for identifying, predicting, evaluating and mitigating the impacts on environment due to any developmental project or activities
- Learn how to prepare an impact assessment report and devise an environment management plan

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Gain knowledge of various pollution sources and their impacts	14	Understanding
CO2	Appreciate the need for minimizing the environmental impacts of developmental activities	12	Understanding
CO3	Apply various methodologies for assessing the environmental impacts of any developmental activity	14	Analysing & Evaluating
CO4	conduct an environmental audit	18	Analysing & Evaluating
	Series Test	2	

CO – PO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2			1			2
CO2						2	2
CO3	2			3	2		3
CO4				2	1		2

3-Strongly mapped

2-Moderately mapped

1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Gain knowledge of various pollution sources and their impacts		
M1.01	Understand different types of Pollution and Pollutants	2	Understanding
M1.02	Understand air pollution, pollutants and their impacts	3	Understanding
M1.03	Understand water pollution, pollutants and their impacts	3	Understanding
M1.04	Understand noise pollution, pollutants and their impacts	2	Understanding
M1.05	Identify the scale of impact of local and global pollutants	4	Understanding
Contents: Different types of pollution and pollutants: Classification of Pollution and Pollutants - AIR POLLUTION: Primary and Secondary Types of Pollutants, impacts - WATER POLLUTION: Point and Non-point Source of Pollution, Major Pollutants of Water, Impact of pollutants - NOISE POLLUTION: Sources of Noise, Effects of Noise -- Impacts of pollutants, types, scale of impact-Global, local pollutants. Climate change, Ozone layer depletion, Deforestation, land degradation , Impact of development on vegetation and wild life			
CO2	To appreciate the need for minimizing the environmental impacts of developmental activities		
M2.01	Explain the evolution of EIA in India	4	Understanding
M2.02	Explain why EIA is needed for developmental projects.	4	Understanding
M2.03	Explain a development project impact on environment	4	Understanding
	Series Test – I	1	
Contents: Definition, Need for EIA, Evolution of EIA: Global & Indian scenario - Environmental legislations in India- The Water (Prevention & Control of Pollution) Act 1974, The Air (Prevention & Control of Pollution) Act 1981, The Environmental (Protection) Act 1986- Environmental standards for water, air and noise quality- EIA Notification 2006			

CO3	To apply various methodologies for assessing the environmental impacts of any developmental activity		
M3.01	Understand environmental clearance process in India	5	Analysing & Evaluating
M3.02	Prepare an environmental impact assessment report	4	Analysing & Evaluating
M3.03	Identify the various methodologies for impact assessment	5	Analysing & Evaluating
Contents: Environmental clearance process in India: Screening, Scoping, Public Consultation, Appraisal-Form1-Category of projects-Generic structure of EIA report-Terms of Reference (ToR) -Types of EIA: strategic, regional, sectoral, project level- Rapid EIA and Comprehensive EIA- Initial Environmental Examination (IEE) EIA methodologies: Ad hoc, checklist, matrix, network and overlay			
CO4	To conduct an environmental audit		
M4.01	Understand the importance of an environmental management plan	4	Analysing & Evaluating
M4.02	Importance of ISO 14001 standards	3	Analysing & Evaluating
M4.03	Explain the need for environmental auditing	5	Analysing & Evaluating
M4.04	Analyse EIA case studies (Indian)	6	Analysing & Evaluating
	Series Test – II	1	
Contents: Environmental Management Plan (EMP): Goal and purpose- Importance of EMP- Content of an EMP- Role of environmental monitoring program Environment Audit: need for audit- audit types and benefits- environmental audit procedure ISO 14001 standards: Importance, salient features - Stages in implementation- Benefits EIA case studies (Indian)- a highway project, a hydroelectric power plant, an airport project			

Text / Reference

T/R	Book Title/Author
T1	Larry W Canter, “Environmental Impact Assessment”, McGraw Hill Inc. , New York, 1995
T2	Betty Bowers Marriott, Environmental Impact Assessment: A Practical Guide, McGraw-Hill Professional, 1997
T3	Environmental Impact Assessment, 2003, Y.Anjaneyulu, B.S Publications
R1	Lawrence, David P., Environmental Impact Assessment (Practical Solutions to Recurrent Problems), Wiley International, New Jersey.

R2	Ministry of Environment & Forests, Govt. of India 2006 EIA Notification
R3	Jain, R.K., Urban, L.V. and Stacey, G.S., Environment Impact Analysis, Von Nostrand Reinhold Company.

Online Resources

S.No	Website Link
1	Prof. Harshit Sosan Lakra, IIT Roorkee, Environmental Impact Assessment [NPTEL] Available: https://onlinecourses.nptel.ac.in/noc23_ar04/preview